Index Calculus for the Elliptic-Curve Discrete Logarithm, Measured:

A Fourteen-Week Programme of Decomposition Oracles, Factor Bases, Gröbner Engines and a Common Cost Harness, with Every Result Placed Against Pollard Rho

Anonymous

Abstract

Index calculus is the method that, on paper, should exploit the most structure in an elliptic-curve discrete-logarithm instance, and in practice has never beaten Pollard rho on a prime-order curve of cryptographic shape. Between July and mid-October 2026 we ran a sustained engineering and measurement programme on this question in one open repository: about 6,900 commits and 450 dated study directories, the index-calculus part of which began in the first week of September. This paper is the record of that programme. We describe the routes we pursued—Frobenius-invariant factor bases on Koblitz curves and their compact-orbit form; SAT and Gröbner-basis decomposition oracles (matrix F4, F5, an inherited F4 and an F6-IC variant); root-finding decomposition oracles; Kummer-tower and symmetry-reduced prime and extension-field systems; isogeny and transfer routes; and a twelve-workstream replication of the 2025–26 literature—the infrastructure that made the results comparable (*ecbench*, a native harness that charges every ECDLP method in one counted unit, group-addition equivalents per $\sqrt{r}$, with sealed records, exact replays, isolation levels and paired cluster-bootstrap intervals), and what improved.

The largest single change came from a suggestion in a manuscript we audited: the roots of Semaev's third summation polynomial in its free variable are $x(R+F)$ and $x(R-F)$, so a two-summand decomposition needs no polynomial system at all. We proved this (also in Lean), generalised it to meet-in-the-middle pair-sum tables for three and four summands, to a CUDA kernel, to Kummer-tower bases and to dense root indices, and built it into the harness. Per decomposition it is 383–1,535× faster than our matrix F4 on Koblitz curves and 3,021–5,802× faster than Buchberger on the five NIST prime curves; end to end it is 50–105× faster than msolve on small prime curves. Inside ecbench it moved the best complete-count Koblitz index-calculus arm at $n = 13$ from $S = 38.8$ to $S = 23.0$, below an XOR-priced F4 at $S = 421$ but still 1.55× the strong signed-Frobenius rho. FGHR's D_4 symmetry cut the $k = 4$ Gaudry solve over $\mathbb{F}_{p^4}$ by 47.5–48.4×, moving a modelled crossover with rho from 2^{150} to 2^{135}; our $\mathbb{F}_p$ F4 reached parity with msolve on square systems; and a compact-orbit Koblitz index calculus with retained precomputation solves one unseen public target at $n = 61$ 13.9× faster online than rho (95% interval 9.8–19.1, 64 targets, pinned host) while costing 83× rho when its set-up is charged. We report the negative results with the same weight: no measured index-calculus arm beats matched rho on the cold, one-target, set-up-inclusive boundary; a group-homomorphic transfer cannot change decomposition counts (we state the conservation lemma); and several literature claims did not survive implementation. Every number cites a committed artefact; nothing here concerns a deployed curve or key recovery.

Contents

1 Introduction **2**

2 Background **3**

3 Measuring index calculus against rho **4**

3.1 The boundary ledger (10 September) 4

3.2 ecbench (from 2 October) 4

3.3 Claim hygiene and change classes 5

3.4 Corrections the accounting forced 6

4 Chronology **6**

5 Root-finding decomposition **7**
5.1 Origin: a suggestion in an audited manuscript . . . 7
5.2 The algorithms . . . 8
5.3 Generalising the observation . . . 8
5.4 Results . . . 8

6 Gröbner-basis engines **10**

7 Koblitz and binary curves **12**
7.1 Frobenius-invariant and compact-orbit bases . . . 12
7.2 The oracle sequence . . . 13
7.3 The one-target online ladder . . . 13
7.4 ECC2K-130 . . . 14

8 Prime and extension fields **14**
8.1 Known-answer ladders over $\mathbb{F}_p$ . . . 14
8.2 PDP oracle programme . . . 14
8.3 Kummer-tower factor bases . . . 14
8.4 Gaudry's $k = 4$ index calculus over $\mathbb{F}_{p^4}$ . . . 14
8.5 Literature claims that did not survive . . . 15

9 Isogeny, transfer and structure routes **15**

10 What improved, and where the frontier stands **16**
10.1 Three frontiers . . . 16
10.2 What would move the cold frontier . . . 18

11 Conclusion **18**

A Organising the publication **19**

B Artefact index **19**

1 Introduction

For a prime-order subgroup $\langle G \rangle$ of size r on an elliptic curve $E/\mathbb{F}_q$, the best generic algorithm for the discrete-logarithm problem is Pollard rho [10, 13], with expected cost $\sqrt{\pi r/(2A)}$ group operations when A automorphisms (negation, and on a Koblitz curve the Frobenius) can be folded into the walk [6, 14]. Index calculus, the method that breaks finite-field discrete logarithms in subexponential time, was brought to elliptic curves by Semaev's summation polynomials [11] and by Gaudry's and Diem's analyses over extension fields [1, 7]. Its open question has always been *where the crossover is*: on which curves, at which sizes and under which accounting does it cost less than rho?

This paper documents fourteen weeks (1 July–14 October 2026) of work on that question in one repository. The index-calculus programme proper ran from 6 September to 14 October; July and August hold the preceding prime-field structure work that motivated it (§4). The output is large: about 6,900 commits, about 450 dated study directories under `research/`, 27 rows of the performance index `benchmarks/README.md`, a boundary ledger (`docs/ic/boundary_targets.json`), a native measurement harness (`src/cryptanalysis/ecbench/`), and a companion manuscript on root-finding decomposition (`papers/rootfinding-decomposition/`). It is easy, in a body of work this size, to improve a headline number by two orders of magnitude and establish nothing; this paper is organised to prevent that. Every improvement is stated with the boundary on which it was measured (oracle stage, complete solve, cold one-target cost, online cost after retained set-up), the unit, the sample, the interval, and the committed artefact that holds the records.

The routes. We pursued six lines, each a section below:

1. *Root-finding decomposition* (§5): replacing the Gröbner solve of the decomposition step by closed-form roots of S_3, then by meet-in-the-middle tables for more summands, then by GPU, tower and dense-index variants.
2. *Gröbner-basis engines* (§6): matrix F4, F5, an inherited F4 that re-uses its root echelon form, an index-calculus-specific F6-IC, an $\mathbb{F}_p$ F4 brought to msolve parity, and an audit that found and fixed one soundness bug.
3. *Koblitz and binary curves* (§7): Frobenius-invariant factor bases [5], their compact signed-orbit form, SAT oracles [12], and a ladder of one-target online comparisons with rho from $n = 53$ to $n = 83$.
4. *Prime and extension fields* (§8): $j = 0$ and $a = -3$ known-answer ladders, the Kummer-tower factor base of Petit–Kosters–Messeng [9], and Gaudry's $k = 4$ index calculus over $\mathbb{F}_{p^4}$ with the D_4 symmetry of Faugère–Gaudry–Huot–Renault [4].
5. *Isogeny, transfer and structure routes* (§9): conductor gaps in the ECC2K-130 isogeny class, isogeny fibres, transfer-aware index calculus, GHS and Weil descent; mostly negative, with one lemma that explains why.
6. *Literature replication* (§8, §9): the *lit2026* programme, which turned each finding of a 2025–26 literature search into a measured change, a diagnostic, or a recorded negative.

Underneath all six sits the measurement work of §3: a single harness (*ecbench*) that charges rho, BSGS, the kangaroo and index calculus in one unit on one-target workloads, and the accounting corrections that the harness forced on earlier claims.

Contributions.

- A unit and a boundary: S = group-addition equivalents/$\sqrt{r}$, cold one-target with every phase charged, beside a separately reported online window; sealed, replayable records; and a frontier definition under which two numbers are comparable only when every component of the measurement domain matches (§3).
- Root-finding decomposition taken from an observation to an engineered, verified, harness-integrated oracle family, with a machine-checked proof of the identity it rests on (§5).
- A documented map of which levers move a constant, which move nothing, and which make things worse, across about 40 measured variants of the decomposition, factor-base and linear-algebra stages (Table 2).
- An honest frontier (§10): the cold one-target frontier is still rho; the retained-set-up online frontier has an admitted index-calculus point; the stage frontier moved substantially.

What this paper does not claim. No asymptotic sub-rho algorithm, no change to the Gaudry–Diem picture for prime fields, no statement about ECC2K-130, P-256 or any deployed curve beyond labelled arithmetic on public parameters, no key recovery. Every instance is a public synthetic or standard-parameter known-answer instance. Every change is classed, in the repository's vocabulary, as an ADVANCE (an exponent or a crossover moves under complete accounting), ENGINEERING (a constant moves), RELABELLING, or ACCOUNTING; with one modelled exception (§8.4) every improvement in this paper is ENGINEERING or ACCOUNTING.

How to read the numbers. "$X\times$" always means baseline cost divided by candidate cost on the stated boundary. Stage ratios (one decomposition, one Gröbner call) are never presented as end-to-end speed-ups; wall-time ratios from hosts that did not earn an isolation level are labelled *descriptive*; a cost with unpriced work is labelled a *lower bound*. Appendix B lists the artefact behind each table.

2 Background

Rho and its floor. With A efficiently computable automorphisms acting on $\langle G\rangle$, a rho walk on orbit representatives has expected cost $\sqrt{\pi r/(2A)}$ group additions. $A = 2$ (negation) on every curve; $A = 2n$ on a Koblitz curve $y^2 + xy = x^3 + ax^2 + 1$ over $\mathbb{F}_{2^n}$, where the Frobenius $\pi(x, y) = (x^2, y^2)$ acts on $\langle G\rangle$ as a scalar λ [6, 14]. ECC2K-130 is the case $a = 0$, $n = 131$ of this family; its record GPU rho attempt is a separate paper in this repository (`paper/ecc2k130/`).

Index calculus on elliptic curves. Fix a factor base $\mathcal{F} \subset E(\mathbb{F}_q)$. A *relation* is a decomposition $aG + bQ = R = P_1 + \cdots + P_m$ with $P_i \in \mathcal{F}$; once about $|\mathcal{F}|$ relations are found, linear algebra over $\mathbb{Z}/r$

yields $\log_G Q$. The pipeline has five stages (factor base, decomposition oracle, relation yield, rank/linear algebra, verification) and a sixth, the comparison with rho on the same target. The *decomposition oracle* decides whether R splits over $\mathcal{F}$. Semaev [11] reduced it to the roots of the summation polynomial $S_{m+1}(x_1, \ldots, x_m, x_R)$, which vanishes exactly when $\pm P_1 \pm \cdots \pm P_m = R$ for points with those abscissae. Over $\mathbb{F}_{2^n}$ or $\mathbb{F}_{p^k}$ the factor base is chosen as the points whose abscissa lies in a subspace or subfield, and Weil restriction turns $S_{m+1} = 0$ into a polynomial system over the base field; the textbook oracle solves that system with a Gröbner-basis engine (F4/F5, [2, 3]) or a SAT solver [12]. Gaudry [7] and Diem [1] give the asymptotics over extension fields; for prime fields no factor base is known that makes the method competitive, and summation-polynomial systems over $\mathbb{F}_p$ are the subject of the Kummer-tower and rational-map bases of Petit–Kosters–Messeng [9].

Frobenius-invariant bases. Galbraith, Granger, Merz and Petit [5] observe that a factor base with $\pi(\mathcal{F}) = \mathcal{F}$ splits into Frobenius orbits of size dividing n, with $\log \pi^k(P) = \lambda^k \log P$. The unknowns collapse by $\approx n$, the matrix by $\approx n^2$, and ordered decompositions save $m!$—more than rho's $\sqrt{2n}$, but from a base too far behind to overtake it at deployed sizes. Invariant bases come from linearised polynomials attached to the factors of $x^n - 1$ over $\mathbb{F}_2$; the subspace dimension is $\ell = \mathrm{ord}_n(2)$, so the base size is fixed by n (Table 1 of `RESEARCH_KOBLITZ_INDEX_CALCULUS.md`).

Symmetries of the system. The decomposition system is invariant under the symmetric group S_m; rational 2-torsion adds sign changes, and Faugère, Gaudry, Huot and Renault [4] show that in the coordinate Y with $Y(P+T) = -Y(P)$ the invariance group becomes the Weyl group D_m, with a much smaller quotient. We measure this at $m = 4$ in §8.4.

3 Measuring index calculus against rho

The programme's first lesson was that index-calculus numbers from different studies did not compose: one study timed a decomposition, another a relation, a third a whole multi-target batch with shared tables, each on a different host and in a different unit. Four mechanisms, built in this order, fixed that.

3.1 The boundary ledger (10 September)

`docs/ic/BOUNDARY_TARGETS.md` and its machine twin `docs/ic/boundary_targets.json` give each regime (Koblitz, binary, prime) and each pipeline stage (`factor_base`, `decomposition`, `relation_yield`, `rank`, `end_to_end_dlp`, `vs_rho`) a current record, a next target and acceptance gates. A row promotes only when its evidence JSON carries every field the measurement schema requires for that stage—for example, a `decomposition` claim that beats an algebraic frontier must report the first-fall degree, and a `vs_rho` claim needs one target, the identical public point for both methods, verified scalars, five exclusive IC online phases, the automorphism discount, matched resources and a host-isolation receipt. Missing fields fail closed. The primary `vs_rho` question is the online cost of *one* previously unseen target after reusable IC preparation; multi-target and amortised results are secondary and must be declared as such.

3.2 ecbench (from 2 October)

ecbench (`src/cryptanalysis/ecbench/`, `src/bin/ecbench.rs`, documented in `docs/ecbench/README.md`) runs every ECDLP method behind one interface, on the same single-target workloads, in one unit, into sealed records.

Unit.

$$S = \frac{\text{total group-addition equivalents (GAE)}}{\sqrt{r}}, \qquad S_{\text{floor}} = \sqrt{\pi/(2A)}.$$

An addition or doubling is one GAE; a scalar multiplication is charged as the additions it performs; native work (lookups, row operations, square roots, Artin–Schreier solves, word XORs) is priced at a pinned conversion where the repository has one and is otherwise recorded as *uncharged*, which marks the record's S as a *lower bound*. Plain rho has $S \approx 1.25$.

What a measurement is. One workload is one curve, subgroup, generator and target; nothing survives between executions, so every table a method builds is inside its cost (*cold*). Beside the cold S every record carries the one-target *online window*, split for index calculus into the five exclusive phases `target_query`, `target_PDP`, `target_relation_check`, `target_descent`, `target_recovery_check`. Targets are either planted or *public*—hashed to the subgroup so that nobody knows the logarithm—and the runner verifies $[k]G = Q$ in its own process; `ecbench verify` checks it again from the files. Failed, exhausted and timed-out runs are rows, never dropped.

Identity, sealing and replay. Every object (curve, factor base, method parameters, spec, session, host environment class) is named by a hash of what it is (`ECBS1h...` for sessions, `ECBND1h...` for bounds). Each execution appends one sealed record; `ecbench verify` replays measured records and emits an audit receipt whose SHA-256 is the replay certificate. Operation counts are deterministic, so a replay must match to the last digit; a mismatch is preserved as a failed audit (§3.4).

Isolation. A run earns L0 (recorded under a host capsule), L1 (pinned, affinity read back, no foreign SMT sibling, CPU/wall ≤ 1.05), L2 (quiet: zero steal, foreign busy time $\leq 1\%$, run-queue delay $\leq 0.5\%$, ≤ 50 preemptions/s) or L3 (isolated cores, fixed governor, bare metal). Wall time is admitted only at the spec's required level (default L2) with at least five admitted pairs and an A/A interval; otherwise it is printed as *descriptive*. Operation counts need no isolation. macOS hosts earn L0; a GitHub-hosted runner earned L1 on 72/72 runs and matched a macOS session's operation counts to the last digit. *isolab* prepares lab hosts for L3 and cross-host replay.

Statistics. The comparable statistic is the mean S (the floor is an expectation). Comparisons are paired by (workload, round) with shared seeds, reported as a ratio of totals $\sum S_B / \sum S_A$, never a mean of ratios, with a two-stage cluster bootstrap that resamples workloads, then pairs. An operations A/A is exactly 1 by construction and checks determinism. A claim about scaling fits exponents over at least four sizes.

Bounds and the frontier. `ecbench` also records sealed *bounds*—constant and exponent with intervals, per phase—and a challenge protocol whose verdict says whether a candidate advances, trades, matches or regresses an incumbent and at which level (exponent, constant, primitive). The incumbent for cold one-target Koblitz operations is `rho.signed_frobenius_strong` (bound `ECBND1h05a2ea918f3c`): $1.089\times$ the floor $[0.927, 1.261]$, fitted exponent $\alpha = 0.451$ $[0.399, 0.512]$, 144 runs at six sizes (`research/ecbench_frontier_20261013/`).

Definition 1 (comparability)**.** A performance point is keyed by the tuple

$$D = \big(\text{curve and subgroup, target law, cold} \mid \text{online} \mid \text{stage, unit and unpriced work,}\\ \text{set-up amortisation } q\text{, memory envelope, correctness/replay/isolation class}\big).$$

Two points are ratio-comparable only when every component of D except the method matches; a method is on the frontier of a domain when no admitted point is no worse on every axis and strictly better on one.

This rules out the four collapses we saw most often: online wall as cold cost; word XORs, probes, field operations and GAE as interchangeable; a lower bound as a complete cost; an L0 wall ratio as an admitted speed-up.

3.3 Claim hygiene and change classes

Every study directory states a protocol before it runs (frozen inputs are committed first), keeps failures, and closes with a decision in one of the four classes of §1. A controlled-experiment rule requires a negative control from the right null population, a positive control that proves the detector can see the effect, and the same instrument on every arm before any comparative statement. Blockers are recorded as named problems (`problems/`) with exit criteria rather than silently dropped. Measurement code is native (Rust); Python only lays out recorded values.

3.4 Corrections the accounting forced

Building the harness and the ledger exposed errors in earlier claims, including our own. We list them because they change how earlier numbers must be read.

1. *Online Koblitz ratios relabelled* (7–8 October, `docs/ic/PLAN_IC_ACCOUNTING_FIXES_20261007.md`). Same-point online ratios of 124.85× ($n = 61$), 2,864.1× ($n = 71$), 1,189.7× ($n = 73$) and 8.2× ($n = 83$) had verified answers but no host-isolation receipt; they are *exploratory*. Only the later 64-target pinned panel at $n = 61$ is admitted (§7).
2. *ECC2K-130 extrapolation constants corrected* (`RESEARCH_ECC2K130_IC_FEASIBILITY.md`): the structural model predicts $C = 0.25$ in probes/relation $\approx C\,r/(n^2K^2)$; the fitted C is 0.27 at $n = 61$ but 1.2–1.5 at $n = 71, 73$, an unexplained gap now stated rather than absorbed.
3. *"F4" that was Buchberger* (`research/ecbench_rootfinding_assessment_20261012/REPORT.md`): the NIST benchmark of the root-finding manuscript printed its baseline as F4 but called `groebner_f4::buchberger`; its 3,021–5,802× are ratios to Buchberger. The manuscript's claim that no implementation can use fewer than linear field operations per ideal is too broad once a reusable pair table is allowed.
4. *Unpriced word XORs* (`research/ecbench_rootfind_faster_20261014/`). Inherited F4's Boolean word XORs were counted but not charged, so its $S = 16.7$ at Koblitz $n = 13$ was a lower bound that looked competitive with rho (14.4). Pricing them gives $S = 421.4$.
5. *Cross-host counter drift* (`research/ecbench_frontier_20261013/audits/`): a replay-all audit of the root-finder session on x86_64 matched 120/168 records; all 48 inherited-F4 rows differed in solver word-XOR counts (e.g. 301,961 versus 300,531), with every scalar, trial count and charged GAE unchanged. The failed receipt is kept; the F4 stage comparison is historical until the drift is explained.
6. *A soundness bug* (`docs/ic/GROEBNER_F4F5F6_AUDIT_20261010.md` §2.1): a degree-bounded `f4_fp::solve` could certify solutions of an inconsistent system (over $\mathbb{F}_5$, $\{x^3 + 1, xy - 1, x + y\}$ returned two "solutions"). Fixed by evaluating every candidate on the input.
7. *F6-IC charged cost* (§6): a stage win in word XORs became a 64.6–1,353× loss in charged S once its batched geometry was counted—recorded as an accounting result, not an advance.

4 Chronology

Figure 1 shows commit activity on `main` by ISO week. Theme lines are case-insensitive keyword matches on commit subjects (a commit may match several themes or none; the regular expressions are in the paper's `README.md`), so they indicate where effort went, not how much of it was index calculus. The grey bars include unrelated work in the same repository (the ECC2K-130 GPU rho campaign, an isogeny KEM, post-quantum benchmarks), which dominates week 37.

Prelude: July and August. The prime-field *ecdlp autolab* ran numbered threads on curve structure—universal order-2 Frobenius forms, the CM-73 family $\{19, 37, 79, 109\}$, sextic-twist pairs that are Howe-glueable (5 of 15)—and its draft "An Empirical Resistance Map for Prime-Field ECDLP" (`research/ecdlp_autolab/PAPER_DRAFT.md`). Those threads found structure but no solver advantage, which is what turned the programme towards index calculus, where structure is supposed to pay.

Phase 1, 6–14 September: Koblitz index calculus from scratch. The ECC2K-130 rho client and an audit of a public index-calculus benchmark's SAT/UNSAT labels (6 Sept) were followed by Frobenius-invariant factor bases with a matrix-F4 Semaev oracle, an XOR-native SAT oracle after Trimoska et al. (7 Sept), and a $\sim 50\times$ faster Semaev solve the same day. On 8 September a scaling experiment showed the SAT route does not scale and that the refutation frontier belongs to F4; pairs-and-solve replaced triple enumeration for S_4. The boundary ledger was codified on 10 September, together with a meet-in-the-middle oracle, incremental modular linear algebra and a CADO-style split into factor-base logs then descent. A burst of more than 100 numbered pipeline stages (12–13 Sept: PCLMUL field products, fused pair collection, scalar-blind targets, rank-aware scans) took the multi-target $n = 53$ pipeline to a 1,024-target process-wall crossover against batched rho. On 13–14 September six pre-registered ECC2K-130 decomposition experiments ran ("exists, admissible, not findable"; no falsifier fired), and the repository began to require frozen benchmarks and full-pipeline evidence for any ECDLP speed claim.

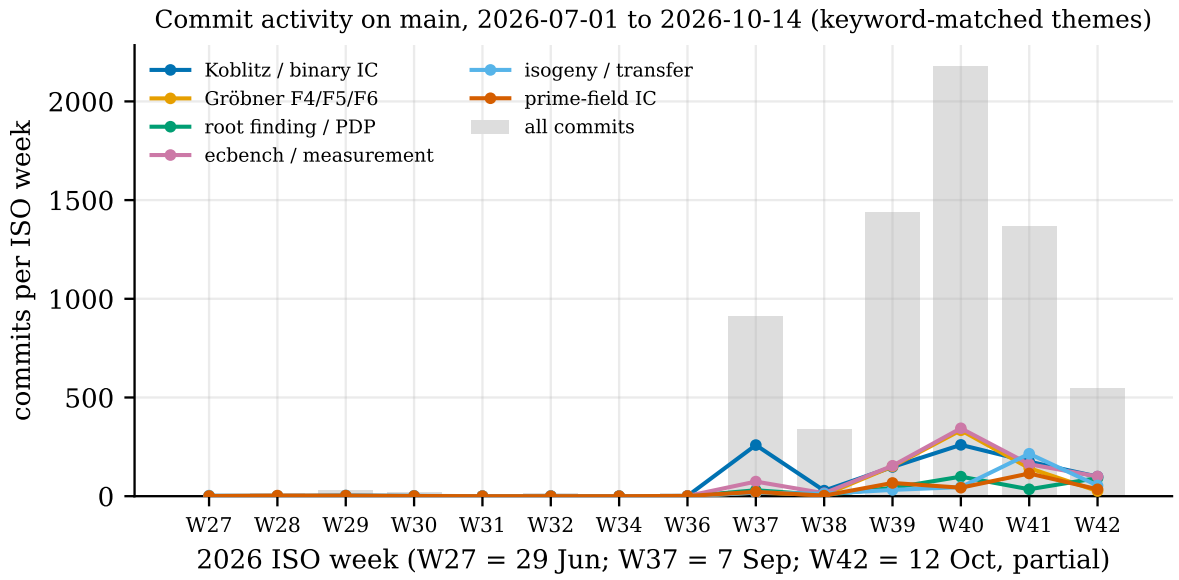


Figure 1: Commits per ISO week on `main`, 1 July–14 October 2026 (6,882 commits; week 42 partial). Lines: commits whose subject matches each theme's keywords. Source: `git log` at the paper's base commit; values in `figures/data.json`.

Phase 2, 15 September–1 October: engines and the compact orbit. Inherited F4 (22 Sept) became the default algebraic oracle; a fast binary-field hot path, Itoh–Tsujii inversion and a parallel radix-sorted pair-table build landed (23–24 Sept); a long Boolean F4/F5 kernel series (AVX2/AVX-512 unpackers, compact columns, pivot orders; 26 Sept–3 Oct) ran in parallel. The compact signed-Frobenius orbit producer beat batched rho on multi-target $n = 61$ batches (25–26 Sept), and from 2 October carried the one-target online ladder to $n = 71$ and (6 Oct) $n = 73$ on 128-bit words, after $n = 63$ was closed as non-viable. A fixed-policy cold comparison on orbit-disjoint public points (1 Oct) was a measured no-go.

Phase 3, 2–10 October: one harness, and the audit it forced. *ecbench* landed on 2 October. F6-IC and a 23-study $n = 83$ campaign (4–6 Oct), the registered $m = 3$ refutation-degree law (5–11 Oct), the ECC2K-130 feasibility extrapolation (5 Oct), the accounting correction plan, conductor-gap and novel-direction notes (7 Oct), and the transfer-aware, isogeny-fibre and Gröbner-engine audits (10 Oct) followed.

Phase 4, 11–14 October: literature, root finding, consolidation. The lit2026 programme (11 Oct) turned a literature search into twelve workstreams. A supplied manuscript on root-finding decomposition was frozen, audited and re-measured in ecbench (12 Oct), and the identity it rests on was generalised across PDP oracles, a CUDA kernel, Kummer towers, Koblitz engines and dense root indices (12–14 Oct). The admitted $n = 61$ online panel ran on two hosts (12 Oct). Gaudry's $k = 4$ solve was made generic and given the D_4 symmetry; the frontier synthesis and its replay audit closed the period (13 Oct); the last root-finding loops priced F4's word XORs and ran out of levers against rho (14 Oct).

5 Root-finding decomposition

5.1 Origin: a suggestion in an audited manuscript

The textbook decomposition oracle builds Semaev's polynomial system and solves it with a Gröbner-basis engine. On 12 October a 13-page manuscript, *Point Decomposition by Root Finding*, was supplied for assessment. We froze it (`research/ecbench_rootfinding_assessment_20261012/source-paper.pdf`, SHA-256 `743d3d3e...7c01`) and audited it against its source branch before measuring anything. Its central suggestion is simple: for two summands, fixing one factor-base point turns S_3 into a quadratic in the remaining variable whose roots are already known.

Algorithm 1 Prime two-summand oracle by direct subtraction

Require: target R, base $\mathcal{F} = \{F_1, \ldots, F_N\}$, hash set H of abscissae (or full points) of $\mathcal{F}$

$d_i \leftarrow x_R - x_{F_i}$ for all i; invert all d_i with one field inversion and $3(N-1)$ multiplications (Montgomery's trick)
for $i = 1..N$ **do**
  for $\varepsilon \in \{+1, -1\}$ **do**
    $\lambda \leftarrow (y_R + \varepsilon y_{F_i})\, d_i^{-1}$; $\quad x_3 \leftarrow \lambda^2 - x_R - x_{F_i}$ ▷ $x(R - \varepsilon F_i)$
    if $x_3 \in H$ **then** recover the sign of F_j from y_3; emit $R = \varepsilon F_i + \tau F_j$
    end if
  end for
end for

Proposition 1 (roots of S_3)**.** *Let $E : y^2 = x^3 + ax + b$ over a field K of characteristic $\neq 2, 3$, and let $R, F \in E(K)$ be affine with $x_R \neq x_F$. Then*

$$S_3(x_R, x_F, X) = (x_R - x_F)^2\,(X - x_+)(X - x_-), \qquad x_\pm = x(R \pm F),$$

so $S_3(x_R, x_F, X) = 0$ iff $X \in \{x(R+F), x(R-F)\}$.

Proof sketch. With $\lambda_\pm = (\pm y_F - y_R)/(x_F - x_R)$ the chord slopes, $x_\pm = \lambda_\pm^2 - x_R - x_F$. Expanding $(X - x_+)(X - x_-)$ and eliminating y_R^2, y_F^2 with the curve equation gives exactly Semaev's $S_3/(x_R - x_F)^2$. The identity is machine-checked over an arbitrary field in Lean 4 with Mathlib (`papers/rootfinding-decomposition/lean/S3Roots.lean`, theorems `SemaevS3.S3_factor` and `SemaevS3.S3_root_iff`; no `sorry`; axioms `propext`, `Classical.choice`, `Quot.sound`). □

Hence the two-summand system has a parametrised variety: $R = \pm F_i \pm F_j$ iff $x(R \mp F_i) \in x(\mathcal{F})$. No polynomial system is solved; the oracle is one subtraction and one table probe per base point.

What the audit kept and corrected. The identity is exact under the stated generic affine condition, and a returned witness can be re-added in the group for soundness. Completeness of the exhaustive enumeration holds only with an explicit sign and factor-base convention (the binary table stores full point keys over a base containing both signs; the manuscript's schematic x-only table needed that convention stated). The manuscript's NIST baseline was Buchberger, not F4 (§3.4), and its statement that no implementation can use fewer than linear field operations per ideal fails for repeated queries against a reusable pair table. Larger-summand enumeration has a growing exponent in $|\mathcal{F}|$, and the manuscript did not benchmark it end to end. All of this is recorded in `research/ecbench_rootfinding_assessment_20261012/REPORT.md`.

5.2 The algorithms

Prime fields (Algorithm 1, `src/cryptanalysis/prime_fast.rs`): $\Theta(N)$ field operations and $2N$ lookups per target. *Binary fields* (`src/cryptanalysis/koblitz_index_calculus.rs`, `DecompositionStrategy::PairTable`, made the default): a precomputed table of pair sums $F_i + F_j$ turns an m-decomposition into 1, N or N^2 lookups for $m = 2, 3, 4$—the meet-in-the-middle half of a birthday search, exact and complete like exhaustive search, at the price of $\Theta(N^2)$ set-up and memory. Both are verified fail closed: every decomposition is re-added in the group, S_3 is checked to vanish, Gröbner-variety membership and agreement on refutation are checked against F4, and planted positive controls must be found.

5.3 Generalising the observation

Over three days the identity was carried through every layer of the pipeline. Table 1 lists the steps; each is its own study with a protocol registered before it ran.

5.4 Results

Oracle stage. Per decomposition, root finding is 383× ($\mathbb{F}_{2^9}$, $|\mathcal{F}| = 55$, 254 targets) and 1,535× ($\mathbb{F}_{2^{13}}$, $|\mathcal{F}| = 4{,}005$, 256 targets) faster than matrix F4; 3.5–$4.7 \cdot 10^4$× faster than `groebner_f4` on 16–24-bit

Table 1: Root-finding generalisations, 12–14 October. Every row is ENGINEERING: no slope fell below its generic floor.

Step	What it adds; artefact
Prime subtraction, binary pair table	Prop. 1; batched inversion; $m \leq 4$ tables. `research/semaev_root_vs_groebner_20261012/`
NIST P-192…P-521	standard parameters, 6- and 64-point bases; dedicated P-256 field 11–12× over generic big integers. `research/semaev_rootfind_nist_20261012/`
End to end vs msolve	complete DLPs with only the oracle swapped. `research/semaev_rootfind_e2e_20261012/`
PDP oracle family	`rf-mitm`, `mitm3`, `mitm4` (2-sum table); `rf-pairs`, `pairs3` (pairs then solve by the group law); `subtract2`; binary rungs $\ell = 8..14$, prime Kummer towers $m = 2, 3, 4$. `research/pdp_root_finder_20261012/`
CUDA kernel	one thread per (target, slice); per-thread Montgomery inversion; SoA base; shared-memory hash for $\lvert\mathcal{F}\rvert \leq 2048$; 64-bit and P-256 fields. Validated in host emulation only (no GPU available). `RESEARCH_GPU_ROOTFIND_DECOMPOSITION.md`
ecbench engines	`pdp3-koblitz:m=2|3,engine=rootfind`, `rootfind-subtract`, later `rootfind-window` and `rootfind-negfold`. `research/ecbench_rootfind_engine_20261012/` and successors
Kummer-tower base	PKM's μ_{2^t} base with the root-finding oracles, 20–28-bit primes. `research/pkm_tower_ecbench_20261013/`
Dense root index, $n = 83$	packed five-byte S_3 root table, prefix widths K1271–K1798, 8 GiB of slots. `research/n83_dense_root_20261013/`
Prime tuning	pair-table size, arity, batched table inversions, synchronised lanes. `research/ecbench_rootfinding_\{tuning,arity,batchtable,lanes\}_2026101*/`

prime curves ($|\mathcal{F}| = 6$, 40 targets each); and 3,021–5,802× faster than Buchberger on the decomposition ideal of the five NIST curves (Figure 2). Every binary verdict agreed with F4; every prime solution was in F4's variety and every prime refutation was F4's unit ideal. In the PDP study, `rf-mitm` refutes a binary target in 3.1 μs at $\ell = 8$ and 482 μs at $\ell = 14$, against the PS-0 baseline's 9.3 ms and 55.5 s (3,007–115,160×), costs $5|\mathcal{F}| - 3$ multiplications and $2|\mathcal{F}|$ lookups a target, and sits on the pair-table floor at 1.000 ± 0.027 bits per rung; at prime $m = 3$, $N = 18$ it uses 197 multiplications where F4 uses $8.99 \cdot 10^{13}$ multiply-adds; 707/707 brute-force agreements (`research/pdp_root_finder_20261012/`).

Complete solves. End to end, the gain is two orders of magnitude smaller and not one-sided: 50–105× against msolve on 16–24-bit prime curves; 122× on the one binary instance ($n = 23$) large enough to amortise the pair table; and F4 *faster* at $n = 5, 9, 13$, where a solve needs too few decompositions to pay for the table. Inside ecbench, with every group operation charged, the root-finding IC beats inherited F4 in solve wall by about 350× at $n = 13$ (admitted) and about 10^4× at $n = 23$ (descriptive), yet stays 2.9–25,116× the strong rho in S on binary curves and 29–141× rho on prime curves, with the gap widening (`research/pdp_root_finder_ecbench_20261013/`).

The Koblitz loops. Figure 3 follows the best complete-count root-finding arm at Koblitz E_0, $n = 13$ through four registered improvement loops (8 public targets × 3 rounds each, L0 operation counts). The first engine had $S = 38.82$ ($m = 3$) and 28.38 ($m = 2$). A windowed third-summand scan cut $m = 3$ by a paired 0.828 $[0.783, 0.863]$; a fold over negation (`rootfind-negfold`) took $m = 2$ to 23.22 and then 22.97. Two further cycles of denser and sparser bases, windowed negfold and plain root finding were dead ends (`research/ecbench_rootfind_vs_rho_20261014/`). With F4's word XORs priced, the paired ratio negfold/F4 is 0.0545 $[0.047, 0.064]$, a complete-against-complete comparison; against strong signed-Frobenius rho ($S = 14.82$) the best complete arm is still 1.55×. A compact-orbit base with a counted Frobenius pair table reaches $S = 15.55$, below the then-unpriced F4 (16.75), but it is itself a lower bound (factor-base construction natives unpriced), and it is 1.08× rho. Amortised over 64 targets that arm reaches $S \approx 11.2$, below rho; by the ledger's rules that is a secondary, multi-target result, not a cold claim.

Prime tuning in ecbench. On a 20-bit prime curve, the pair table's counted S is 0.018861 $[0.018547, 0.019242]$ of direct subtraction (24 pairs; prime native work unpriced, so a bounded ratio), and 0.057047 $[0.041159, 0.092069]$ on a 17-bit Koblitz curve, fully priced, where the table set-up is about

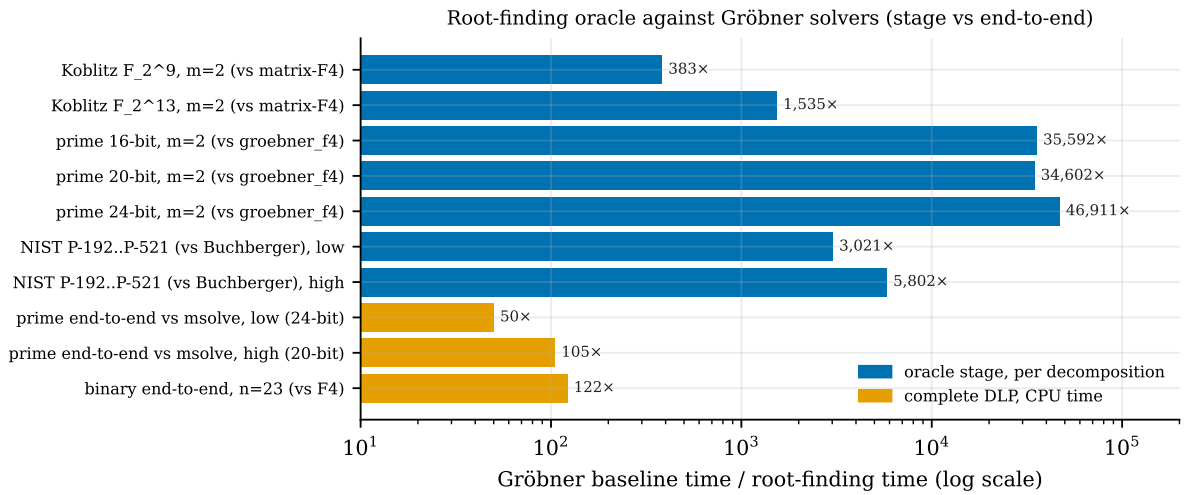


Figure 2: Root finding against Gröbner solvers on identical targets. Blue: one decomposition (oracle stage only). Orange: complete discrete-log computations with the oracle swapped and everything else fixed (prime: CPU time against msolve, 20/24-bit totals composed from sampled per-query cost; binary: one cold solve including table build). Sources: `RESEARCH_SEMAEV_ROOTFIND_VS_GROEBNER.md`, `papers/rootfinding-decomposition/README.md`.

68% of the arm's cold cost. Doubling the table from 32 to 64 abscissae gives 0.830 $[0.697, 0.962]$; three summands instead of two costs $1.431\times$ $[1.225, 1.716]$; batching the table inversions leaves counted GAE unchanged (1.000; 4,096 versus 64 inversions, set-up wall 243 versus 128 μs, descriptive); and a train-set lane selection did not resolve a hold-out winner (0.979 $[0.919, 1.038]$; 1,088/1,088 verified executions) (`benchmarks/README.md`, rows of 12–14 Oct).

Dense root index at $n = 83$. Widening the prefix base from K1500 to K1798 cut mean probes per relation from 300,021.9 to 169,069.8 (paired difference 130,952, 95% [52,058, 209,991]) and gave a one-point online ratio of 18.528 over same-point rho (IC 63.3 ms, rho 1,173.3 ms), with 339.0 s of reusable preparation and an 8 GiB table outside the window. The host was shared, so the result is exploratory; two K1500 runs hit the 900 s timeout and are retained.

Reading. Root finding is the largest constant-factor change of the programme at the oracle stage, and the one with a proof behind it. It does not change the asymptotic count of index calculus: its cost is $\Theta(|\mathcal{F}|^{m-1})$, every measurement is at $m \leq 4$, and it leaves relation yield, factor-base size and linear algebra untouched. Its effect on the cold frontier is to move the decomposition stage from dominant to minor, which exposes the stages that now are.

6 Gröbner-basis engines

Before root finding, the decomposition oracle was a Gröbner solve, and for $m \geq 3$ over binary fields it still is. The repository grew a family of engines (`docs/ic/GROEBNER_F4F5F6_AUDIT_20261010.md`, §1; file names below are under `src/cryptanalysis/`):

Matrix F4 and F5 over $\mathbb{F}_2$
(`koblitz_groebner.rs`, `matrix_f5_f2.rs`): Macaulay matrices of the Weil-restricted Boolean system, reduced by a shared Four-Russians kernel (`gf2_elim.rs`, four 8-bit tables, AVX-512 or AVX2).

Inherited F4
(`inherited_f4.rs`, from 22 September, the default algebraic oracle): a DPLL-style tree over partial assignments in which each child *inherits* its parent's echelon form and re-reduces only what the assignment changed, instead of building a fresh Macaulay matrix per node.

F6-IC
(opt-in): inherited F4 plus a factor-base domain gate (a fully assigned summand whose abscissa is not a base point refutes; partial support pruning refutes earlier), and meet-in-the-middle leaves (with

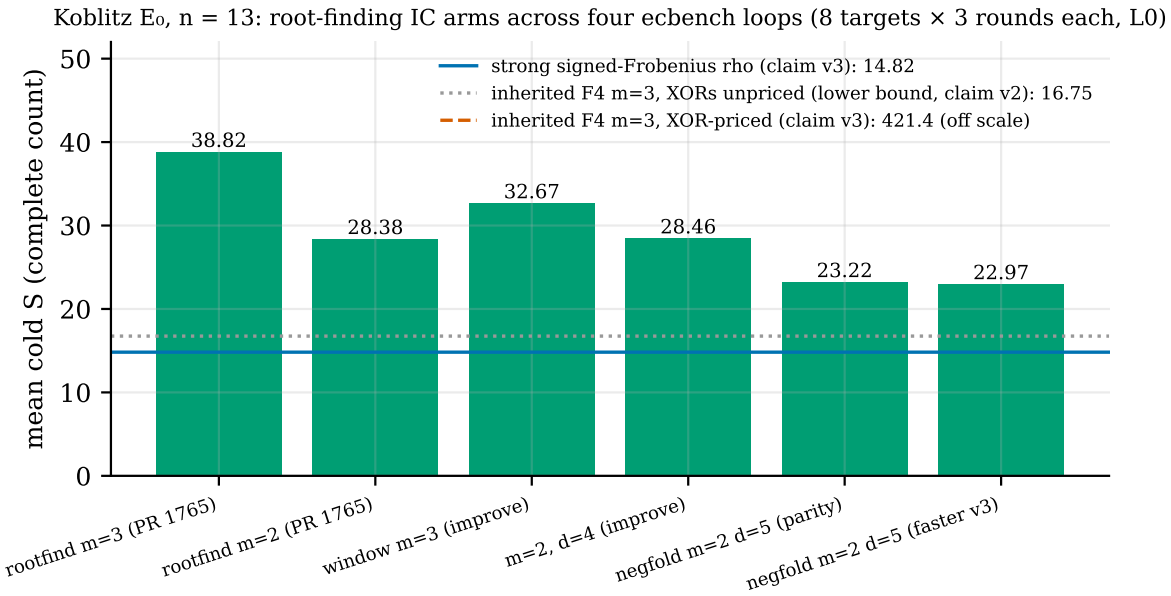


Figure 3: Mean cold S of root-finding IC arms at Koblitz E_0, $n = 13$, $d = 5$ unless stated, across the engine, improve, parity and faster loops. All bars are complete counts. Sources: `research/ecbench_rootfind_engine_20261012/`, `research/ecbench_rootfind_improve_20261013/`, `research/ecbench_rootfind_parity_20261013/`, `research/ecbench_rootfind_faster_20261014/`. Each loop had its own seeds and rho arm (rho 14.12–14.82).

$m - 1$ summands fixed the last is looked up; at $m = 3$ and $|\mathcal{F}| \leq 256$ every compatible second point is added and the third looked up).

$\mathbb{F}_p$ **F4**
(`f4_fp.rs`, `f4_fp_tower.rs`, `sig_fp_tower.rs`): degree-bounded F4 with sparse-by-reducer then dense reduction, a Faugère–Lachartre block for the Kummer-tower oracle, and a signature variant.

Correctness. The audit pinned every engine against every other and against brute force (88+15+9+14+13+12+21+6 passing library tests in the touched modules), and found one soundness bug (§3.4, item 6), two contract defects and several latent holes, all fixed with regression tests.

Engineering results. Figure 4 collects the headline ratios; all are ENGINEERING with identical outputs.

- *Inherited versus matrix F4*: matrix work 198.6× (K13) and 222.1× (random-13) the inherited engine's on identical trees; at $m = 19$ every matrix F4/F5 run timed out at 1,800 s while inherited F4 finished in about 6 s (`research/f4_f5_f6_cross_family_20261011/`).
- *F5 versus F4*: word-XOR ratios 0.6465 / 0.6414 (K13 / random-13), wall 0.79–0.86.
- *F6-IC versus inherited F4*: word XORs about 0.82 at $m = 13$ and 0.77–0.78 at $m = 19$, relation wall 1.7–1.9× faster. But once its batched group additions are charged, F6-IC's S is 64.6–68.8× inherited F4's at $m = 13$ and 1,303–1,353× at $m = 19$: an accounting loss, not a frontier move. At $m = 31$ both engines hit the 3,600 s cap on the first target.
- *Kernel routing and rewrite*: routing every echelon call to the Four-Russians kernel gave 1.75–3.22× on the oracle's own Macaulay matrices; a word-level `pext`/`pdep` rewrite plan made F4 and F6-IC 10–14% faster at $n = 23$; eager rewrite of kept rows beat lazy rows by 1.40–1.62× (`research/inherited_f4_eager_rewrite_20261011/`).
- *Instruction and allocation work*, measured by paired A/B/A′ instruction counts on a frozen eleven-system suite: fused construction and readback, −5.28% (F4) and −11.02% (F5) instructions; native hashing, −3.715% F5 [3.714, 3.726] and −45.1% for the standalone criterion; bounded direct indexing in the F5 packer, −3.947%; row allocation, 42–50% fewer requested bytes. Elapsed time was not claimed for these.
- *Quotient-curve model*: solving the $m = 2$ system directly on the quotient curve instead of a symmetrised control cut F4 word operations by 1.340–1.642× with identical root digests in five interleaved cells (`research/lit2026_w8_summation_models_20261011/`).
- $\mathbb{F}_p$ *F4 to msolve parity* (`research/f4_fp_msolve_parity_20261011/`): Gebauer–Möller criteria,

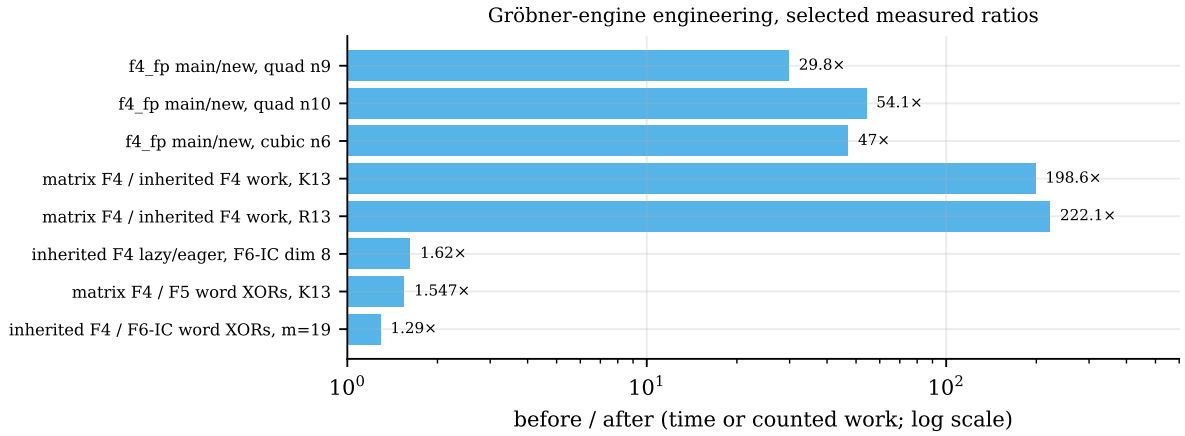


Figure 4: Selected engine ratios (before/after or baseline/candidate): `f4_fp` successive versions (single-threaded wall, L0); matrix versus inherited F4 (counted matrix work); lazy versus eager inherited rewrite (median wall); F4 versus F5 and inherited F4 versus F6-IC (word XORs, stage only). Sources in `figures/data.json`.

packed monomials, interned symbolic preprocessing and a parallel symbolic pass made `f4_fp` 29.8–54.1× faster than its predecessor on square generic systems, with msolve/ours at 0.96–1.22 single-threaded. A day earlier msolve had been 12–24× faster (`research/gb_cas_comparison_20261011/`); every reduced basis agrees with Singular or msolve. On Boolean decomposition systems ($n = 17$, 8 targets) our oracle decides in 7–24 ms per target, while msolve and PolyBoRi spend 40–52 s computing a full basis and Singular exceeds 120 s—a difference of task (decide versus compute the basis), not of engine quality.

What did not work. Degree-2 early-abort probes after a 2025–26 proposal: word-XOR ratios 1.00083–1.00378 (no change) and 0.99714–0.99928 (worse). Gate-first geometry: 1.57–1.78× fewer word XORs but lane-batched F4 roots cost 2.6–4.0× a sequential control. Convolution splitting of the summation polynomial: 1.21–2.07× *more* word operations despite lower degree; Frobenius auxiliary variables: 57–701× more (`research/lit2026_w\{4,5,8\}_*_20261011/`). Symmetry-breaking clauses in SAT did not help (`RESEARCH_KOBLITZ_INDEX_CALCULUS.md`). A registered law for the $m = 3$ refutation degree resolved all 68 units but ended with the verdict "neither": one variant met its generic degree window on 32/32 draws, the other fell below it on 8, and the heaviest degree-10 Macaulay solves were $2.4 \cdot 10^5 \times 2.0 \cdot 10^5$ matrices at 42–51 minutes each (`research/ic_semiregular_law_20261005/`).

7 Koblitz and binary curves

7.1 Frobenius-invariant and compact-orbit bases

The Koblitz implementation (`src/cryptanalysis/koblitz_index_calculus.rs`) builds GGMP's invariant base from the linearised polynomial of a factor of $x^n - 1$, checks $\pi(\mathcal{F}) = \mathcal{F}$ point by point, rewrites each relation into orbit columns with known λ^k weights, and solves the cofactor-corrected system $ha + hbd \equiv \sum \lambda^{k_i} x_{o_i}$. On small fields the orbit collapse is as predicted: $|\mathcal{F}|/\#\text{orbits}$ of 7.86, 10.89, 12.96 at $n = 9, 11, 13$ for relations and 61.7–168.0 for linear algebra, against rho's 4.24–5.10 (`RESEARCH_KOBLITZ_INDEX_CALCULUS.md`). Because $\ell = \mathrm{ord}_n(2)$ fixes the invariant base's size, the programme moved to bases sized to the instance: divisor bases, SAT-optimised bases, a rank-guided base at $n = 53$ (220 orbits, a complete implicit S_5 formula of 22,887 clauses), and finally the *compact signed-Frobenius orbit* base, which stores one representative per orbit and no pair table (`docs/ic/COMPACT_ORBIT_PRODUCER.md`). In the ecbench factor-base suite it is the best measured Koblitz construction: $S = 9.114$ at $n = 17$ (272 points, 8 columns), 9.985× matched signed-Frobenius rho, against full orbit 18.7× and subspace bases 47× (`research/ecbench_frontier_20261013/`).

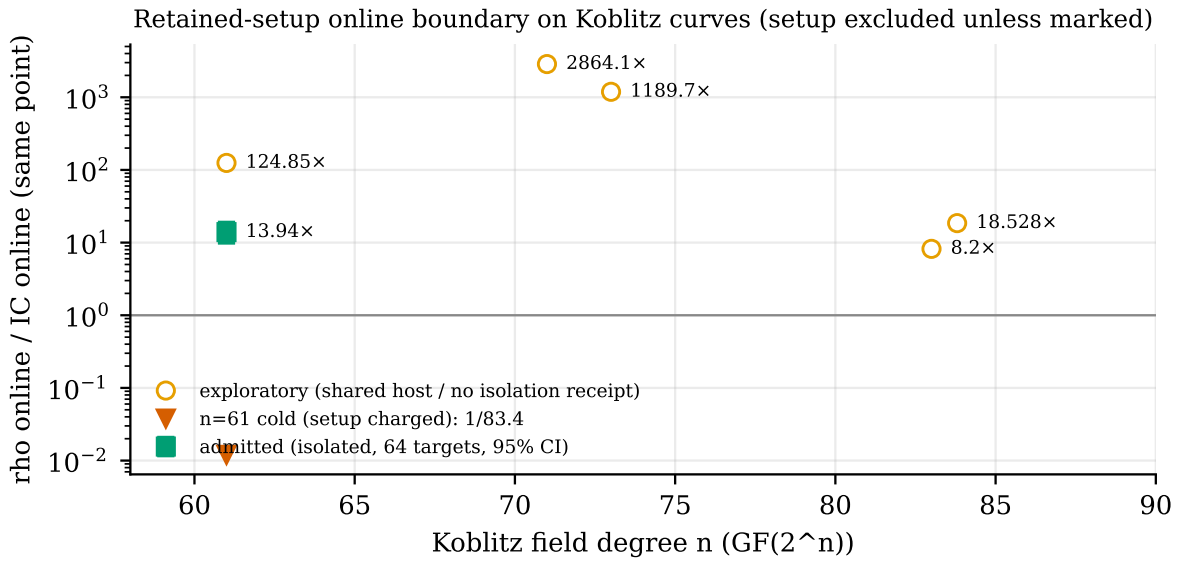


Figure 5: Rho online time over IC online time on the identical public point, reusable IC preparation excluded. Open circles: exploratory (verified answers, no qualifying isolation receipt; three repeats of one point or a single point). Filled square: the admitted 64-workload panel with its 95% bootstrap interval. Triangle: the same $n = 61$ candidate with set-up charged (cold). Sources: `docs/ic/boundary_targets.json`, `docs/ic/PLAN_IC_ACCOUNTING_FIXES_20261007.md`, `research/n83_dense_root_20261013/`.

7.2 The oracle sequence

The binary oracle went through SAT (XOR-native CDCL after Trimoska et al. [12]; decides the $n = 19$, $\ell = 6$ corpus but does not scale), matrix F4 (whose refutation range was the real frontier), pairs-and-solve for S_4 (validated to $n = 21$, $\ell = 7$, still $O(2^{2\ell})$), meet-in-the-middle pair tables, inherited F4 and F6-IC (§6), and root-finding engines (§5). The boundary ledger's current Koblitz decomposition record is an $n = 31$, $m = 2$ F4 cell with an exact block-6 M4RI acceleration; at $n = 53$ an exact Frobenius-orbit S_5 compression removes the coordinate ROM but every unrestricted solver arm remains UNKNOWN.

7.3 The one-target online ladder

The ledger's primary `vs_rho` question is: with reusable IC preparation already done, how long does one unseen public target take, against rho with the automorphism discount on the same point? The ladder (Figure 5):

- $n = 53$: guided shared-log precompute with an explicit pair table (9.34 GB peak RSS); earlier, a 1,024-target shared-log batch beat batched rho in all 15 paired blocks (secondary, multi-target).
- $n = 61$: compact orbit, 600 columns, 73,200 points, 2.73 GB; fresh reproduction 124.85× (exploratory).
- $n = 71$: first rung past 64-bit packing (u128 words), 85,200 points, 2,864.1× median of three repeats of one point (exploratory).
- $n = 73$: 87,600 points, 1,189.7×; the guided-rank precompute took 5.1 h at 57–70 million probes per relation (exploratory).
- $n = 83$, $a = 1$: IC online 5.6–7.9 s against rho 43–71 s, 8.2×; then the dense root index, 18.5× (both exploratory).
- $n = 61$, **admitted** (12 October): $a = 0$, $K = 500$ orbit columns, 61,000 points, 64 fresh public one-target workloads per host, identical candidate manifest on two hosts. Pinned x86_64: median online speed-up 13.94×, 95% [9.81, 19.13], claim-check 64/64; unpinned arm64: 12.87× [7.30, 19.08]. Set-up of about 28 s is outside the window; with it charged, IC costs a median 83.4× rho.

The exploratory ratios are large and irregular because a one-point online window measures the luck of one relation search against one rho walk; the admitted panel's range over 64 workloads is 1.06–648.6×. The admitted point is a constant-factor online result at $n = 61$ (47-bit subgroup) with preparation retained; it is not a cold result, not a scaling law, and not an asymptotic claim. By the frontier's own arithmetic the $n = 83$ preparation would pay for itself after $q^* = 306$ targets (derived what-if: reusable

set-up 339.0 s, online 0.0633 versus 1.173 s; assumes constant per-target cost and zero rho set-up).

7.4 ECC2K-130

ECC2K-130 is the $a = 0$, $n = 131$ member of the same family; the compact producer already solves $n = 41, 53, 61, 71, 73$ of it. Extrapolating the fitted probes-per-relation law $C\,r/(n^2K^2)$ with $C = 1$ gives $1.1 \cdot 10^{18}$ core-years at $K = 600$ and $6.8 \cdot 10^8$ at $K = 10^{12}$ columns (`RESEARCH_ECC2K130_IC_FEASIBILITY.md`, with the unexplained factor-of-5 gap in C of §3.4). The separate GPU rho estimate for the same curve is about 4.3 RTX PRO 6000 card-years (`paper/ecc2k130/`). Index calculus is not a route to ECC2K-130 on this evidence; the isogeny-class routes are discussed in §9.

8 Prime and extension fields

8.1 Known-answer ladders over $\mathbb{F}_p$

Over prime fields the ledger tracks staged known-answer ladders (`docs/ic/boundary_targets.json`, regime `prime`). On $j = 0$ curves the staged driver recovers a 20-bit logarithm ($p = 1{,}048{,}609$) in 56.8–64.9 s of IC wall against 150–153 ms of rho on the identical public point, with split timers and three seeds. On the generic $a = -3$ shape (P-256 and CryptoPro-B class), a Semaev-S_4 three-decomposition family with Cantor–Zassenhaus root finding yields one relation per trial at $p = 65{,}519$ with a 120-point base; at fixed base size, median trials per relation grow from 2 to 28 to 379–506 at 16, 20 and 24 bits, about 16× per four bits. The prime `vs_rho` row is *not achieved* at any measured size.

8.2 PDP oracle programme

The point-decomposition (PDP) speed-up programme (`research/pdp_oracle_speedup_20261011/`) measured the oracle in three steps before root finding replaced it: a baseline (PS-0; 338–452 ns per pair across five rungs, A/A spread $\leq 8.8\%$), a gate-count model of the group-law form (PS-1; 7,410–14,608 fused ternary-logic issues against 70,695–190,275 for the derived baseline, 9.5–13.0×), and a scalar group-law kernel (PS-2; 29.8–40.8 ns per pair, per-pair ratio to baseline 0.067–0.123, 400 corpus root sets agreeing). A split/hybrid solving-degree study then projected its closest relation phase at 22.8× strong rho (`research/pdp_split_dreg_20261011/`), with an initial read-out defect preserved in the record.

8.3 Kummer-tower factor bases

Petit, Kosters and Messeng's base $V = \mu_{2^t}$ (the roots of $x^{2^t} - 1$) gives prime-field summation systems extra structure for an algebraic solver. We ran it end to end in ecbench with the root-finding oracles on five curves of 20–28 bits (896 runs, all verified, all replay audits OK; `research/pkm_tower_ecbench_20261013/`). Index calculus costs 34.0, 41.9, 29.4, 96.6 and 140.6× rho-negation in S (95% intervals in Figure 6), growing by 0.26 ± 0.28 bits of S_{IC}/S_ρ per bit of r, consistent with the $p^{1/6}$ the floors predict. The tower base costs the same as a plain base of the same size to within 1% (ratios 0.999–1.012): PKM's structure helps an algebraic solver, not a table.

8.4 Gaudry's $k = 4$ index calculus over $\mathbb{F}_{p^4}$

Over $\mathbb{F}_{p^k}$ with a base of points whose abscissa lies in $\mathbb{F}_p$, a relation needs a k-decomposition solved by a Macaulay system. We made the `gaudry_quartic` solve generic in the number of unknowns and faster (median whole solve 1.9/4.1/6.4× at $d = 4, 5, 6$, 37.4 → 5.9 s at $d = 6$; the S_5 solve from about 30 minutes to 164–203 s; every count and solution identical). Then we built FGHR's D_4 variant (`research/gaudry_quartic_d4_20261013/`): on a curve with rational 2-torsion T, the coordinate $Y = (x - x_0 - c)/(x - x_0 + c)$ satisfies $Y(P + T) = -Y(P)$, so S_5 in Y is invariant under the Weyl group D_4 (order 192). In the invariants $\sigma_k = e_k(Y^2)$, $\pi = \prod Y_i$, the system has Bézout number 512 instead of 4,096, Macaulay degree 14 instead of 29, and a $13{,}412 \times 11{,}628$ matrix instead of $41{,}780 \times 40{,}920$.

At $p = 269$, the cost C_4 of one decomposition falls from 1.21–$1.23 \cdot 10^{12}$ to 2.54–$2.55 \cdot 10^{10}$ $\mathbb{F}_p$ multiplications, 47.5–48.4×; a solve takes 4.2–5.6 s instead of 164–203 s. Decompositions agree with an independent meet-in-the-middle enumeration on 19 of 19 solved targets; one target in 20 is degenerate (quotient dimension 517) and is discarded, adding about 5% to the relation phase. The Y-line base

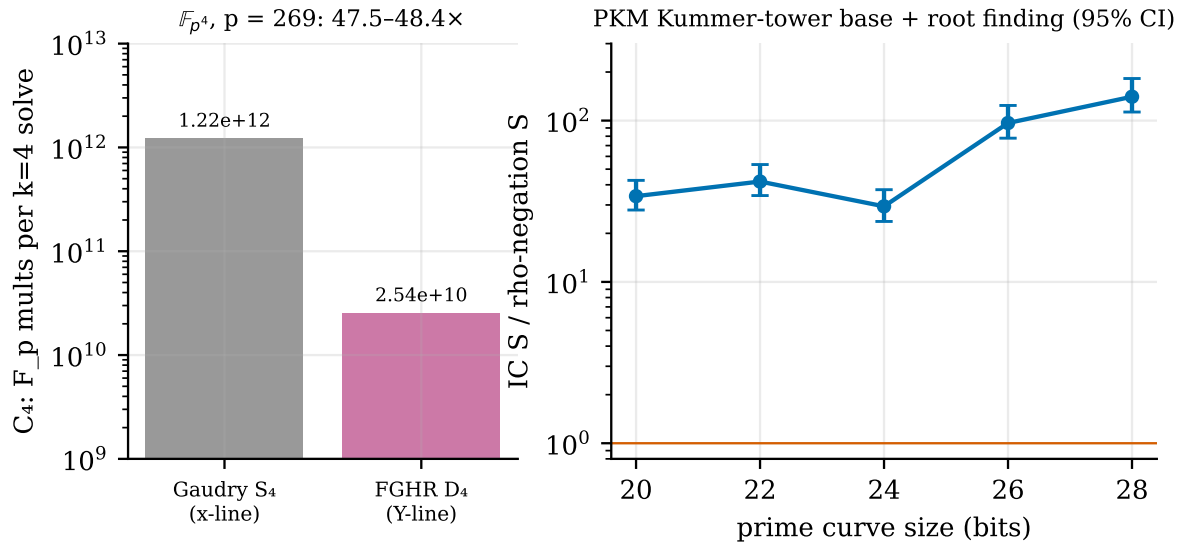


Figure 6: Left: cost C_4 of one $k = 4$ decomposition over $\mathbb{F}_{p^4}$, $p = 269$, Gaudry's S_4 system on the x-line against FGHR's D_4 system on the Y-line (`research/gaudry_quartic_d4_20261013/`). Right: index calculus with the PKM tower base and root finding over rho with negation, eight one-target workloads × three rounds per size, paired 95% cluster-bootstrap intervals (`research/pkm_tower_ecbench_20261013/`).

decomposes about 1/8 as often (pooled ratio 0.133 over 8,000 targets; prediction $1/8 \pm 30\%$ held on one of two curves), but each column covers $\{\pm P, \pm(P+T)\}$, halving the unknowns.

Inserted into the repository's crossover model (`research/notes/index-calculus/RESEARCH_RESIDUAL_WALKS.md` §11.16), the measured C_4 and rate move the $k = 4$ crossover with rho from $2^{149.4}$–$2^{152.7}$ to $2^{135.1}$–$2^{135.8}$ (cofactor $h = 2$, forced by the 2-torsion), and the asymptotic ratio past it from 0.34–0.63 to 0.12–0.22 of rho. This is the one place in the programme where a crossover moves; it moves in an *extrapolated model* ($n^{1/4}$ relations, $n^{1/2}$ linear algebra and rho) fed by a measured constant, so it is classed as ENGINEERING, not an advance. Extension fields of degree four are not the fields deployed curves use.

8.5 Literature claims that did not survive

The lit2026 programme (`research/lit2026_speedup_program/`) ran each finding of a 2025–26 search as a measured change against an interleaved baseline binary. Workstream W12 implemented the zero-minor guess-and-determine attack (Mahalanobis, arXiv 2607.09814v2): it solved 48/48 planted targets on 16–26-bit curves, but its search phases scale with slope 0.905 $[0.769, 1.021]$ in $\log_2 r$ and its cost over rho grows from 260× to 6,715×—exhaustive in r, as the source's own table says. A second flagged ePrint (Hu, 2025/015) did not specify an implementable algorithm and is recorded as blocked. On the generic side, W11 cut BSGS from 16.75 to 3.71 multiplication equivalents per visited point with paired steps (wall 1.15–1.67×), which moves rho's and BSGS's constants, not index calculus.

9 Isogeny, transfer and structure routes

A recurring hope was that some other representation of the same instance—an isogenous curve, a curve at another level of the isogeny volcano, a Jacobian, a Weil descent—would make index calculus cheaper. These routes produced the programme's clearest negatives, and one lemma that explains most of them.

Lemma 1 (transport conservation, `docs/ic/RESEARCH_TRANSFER_AWARE_IC_20261010.md` §1)**.** *Let $\varphi : E \to E'$ be an $\mathbb{F}_q$-rational isogeny (or any homomorphism of rational points), $\kappa = \ker\varphi \cap E(\mathbb{F}_q)$, $R \in E(\mathbb{F}_q)$, $\mathcal{F}' \subset \varphi(E(\mathbb{F}_q))$ a factor base on the codomain and $\mathcal{F} = \varphi^{-1}(\mathcal{F}') \cap E(\mathbb{F}_q)$. Then the number of ordered m-decompositions of R over $\mathcal{F}$ is $|\kappa|^{m-1}$ times the number of ordered m-decompositions of $\varphi(R)$ over $\mathcal{F}'$.*

Proof. Images of decompositions of R are decompositions of $\varphi(R)$. Conversely, for $\sum Q_i = \varphi(R)$ choose

rational preimages P_i of the Q_i; then $\sum P_i - R \in \kappa$, and of the $|\kappa|^m$ choices exactly $|\kappa|^{m-1}$ sum to R (translate P_1 by the kernel element). □

So a homomorphic transfer changes nothing but three curve-specific inputs: the rational kernel symmetry $|\kappa|^{m-1}$ (bounded by the rational torsion; trivial on prime-order curves), the factor-base geometry available on the destination, and the cost of arithmetic there. Sequences of isogenies compose to one homomorphism; a loop is an endomorphism and acts on the subgroup as a known scalar, useful only as a known-log multiplier. Multi-curve factor bases glued by isogenies are single-curve bases in disguise.

Conductor gaps in the ECC2K-130 class. Jao, Miller and Venkatesan [8] prove random self-reducibility among curves with the same endomorphism ring, leaving open whether curves whose conductors differ by a large prime could differ in hardness. The ECC2K-130 class has conductor gap 263; we computed its structure (`docs/ic/RESEARCH_ISOGENY_CONDUCTOR_GAP_20261007.md`) and ran several rounds of experiments. Native counted-S index calculus on two fixed CM pairs gave floor/crater ratios 0.975 (99% $[0.603, 1.451]$) at $\ell = 8209$ and 1.127 $[0.922, 1.342]$ at $\ell = 131101$: no level effect (192 verified records, 96 exact replays; `research/ecbench_conductor_gap_20261012/`). Two-torsion symmetrisation is a quotient isogeny: at a matched base of abscissae it cut F4 row-reduction work 4.1–4.3× ($|H| = 2$) and 17.6–19.6× ($|H| = 4$) over 1,128 systems—but a correction the next day showed that at a matched number of *columns* the saving disappears; it was the cost of a redundant orbit base. A combined Weil descent used 234–529× the F4 operations of the blind system (`research/isogeny_conductor_gap_round3_20261011/`).

Fibre-aware index calculus. Isogeny fibres are torsors whose arithmetic is controlled by one cohomology class (`RESEARCH_ISOGENY_FIBERS_ECDLP.md`); relation generation is isogeny-equivariant at zero lifting cost, and a fibre-closed pair table was the best lever of the conductor-gap rounds (12.8× rho in its best cell). Measured against matched rho across sizes at fixed cofactor, the envelope IC/rho ratio grows from 8.4–8.8 to 35.1–37.0 ($h = 4$) and from 13.4–20.1 to 60.4–63.1 ($h = 72$), as $\#E^{1/6}$: the gap widens, it does not close (816 audited records; `research/fiber_ic_scaling_20261011/`).

Other structure. Symmetrised u-frame bases saved 0.84× at $n = 17$ and 0.76–0.77× at $n = 23$, but cost 4.01× at $n = 31$ (`research/koblitz_symmetrised_e2e_20260927/`). GHS and hyperelliptic covers: ECC2K-130 does map to hyperelliptic curves, but the genus that would pay is missing; end-to-end GHS exists only for $m = 1$ toys (`docs/GHS_SCREEN.md`, ledger implementation caps). The isogeny-loop GLV/GLS work produced a measured 1.09–1.15× faster four-dimensional scalar multiplication (`research/isogeny_loop_glv_20261012/`) and a $\sqrt{\text{élu}}$ kernel cost exponent 0.822 $[0.805, 0.838]$; these speed up curve arithmetic, not the discrete logarithm.

Named obstructions. Blocked requirements are kept open as problem records (`problems/README.md`), among them the complete-cost calibration of root-finding IC (`problems/rootfinding-complete-cost-calibration.md`), with exit criteria rather than abandoned.

10 What improved, and where the frontier stands

Table 2 collects the measured improvements of the programme on the boundary each was measured on. Ratios are baseline/candidate; "stage" means one oracle call or one solver run; "complete" means a verified discrete logarithm with the stated phases.
Derived entries: 53.0 = 1/0.018861; 17.5 = 1/0.057047; 1.21 = 1/0.828 with interval from $[0.783, 0.863]$; 18.3 = 1/0.0545 with interval from $[0.047, 0.064]$; 1.77 = 300,021.9/169,069.8; 1.20 = 1/0.830 with interval from $[0.697, 0.962]$; 1.87 = 18.714/9.985 (one session; the compact arm's S is a lower bound, so the ratio is an upper bound); 1.69 and 25.2 compare means from different sessions with different seeds and are descriptive; F5 and F6-IC ratios are reciprocals of the recorded candidate/baseline word-XOR ratios.

10.1 Three frontiers

Cold, one target, set-up included. Unchanged: generic rho. Every measured index-calculus arm is above its matched rho (Figure 7); the closest complete count is root-finding negfold at 1.55× strong

Table 2: Measured improvements, July–October 2026. Class: E = ENGINEERING, A = ACCOUNTING. Boundary: St = stage, Cx = complete solve outside ecbench, Cold = ecbench cold one-target S, On = online window after retained set-up. Sources: Appendix B.

Change	Bnd.	Cl.	Ratio	Scope
Decomposition oracle: root finding				
S_3 roots, binary pair table vs matrix F4	St	E	383–1,535	$\mathbb{F}_{2^9}$, $\mathbb{F}_{2^{13}}$, $m = 2$
S_3 roots, prime vs `groebner_f4`	St	E	3.5–4.7·10^4	16–24-bit, $\lvert\mathcal{F}\rvert = 6$
S_3 roots vs Buchberger, NIST curves	St	E	3,021–5,802	P-192…P-521
`rf-mitm` vs PS-0 oracle	St	E	3,007–115,160	binary $\ell = 8..14$
Root-finding DLP vs msolve DLP	Cx	E	50–105	16–24-bit prime
Pair-table DLP vs F4 DLP	Cx	E	122	binary $n = 23$ (F4 wins $n \leq 13$)
Pair table vs subtraction	Cold	E	53.0	20-bit prime (bounded)
Frobenius pair table vs subtraction	Cold	E	17.5	Koblitz $n = 17$, priced
Root-finding K13 arm, first to best	Cold	E	1.69	38.82 → 22.97, across sessions
Windowed $m = 3$ scan	Cold	E	1.21	[1.16, 1.28]
Negfold root finding vs XOR-priced F4	Cold	E	18.3	[15.6, 21.3], K13
Dense root index K1798 vs K1500 (probes)	On	E	1.77	$n = 83$, one point
Pair-table size 64 vs 32 abscissae	Cold	E	1.20	[1.04, 1.43]
Gröbner engines				
Inherited vs matrix F4 (matrix work)	St	E	198.6–222.1	K13, R13
F5 vs matrix F4 (word XORs)	St	E	1.55–1.56	K13, R13
F6-IC vs inherited F4 (word XORs)	St	E	1.22–1.30	$m = 13, 19$
`f4_fp` new vs previous	St	E	29.8–54.1	square systems, = msolve
Four-Russians routing	St	E	1.75–3.22	oracle matrices
Eager vs lazy inherited rewrite	St	E	1.40–1.62	dim 8–9
Quotient-curve $m = 2$ model	St	E	1.34–1.64	five cells
Factor bases and symmetry				
Compact orbit vs full orbit	Cold	E	≤ 1.87	Koblitz $n = 17$; compact is LB
FGHR D_4 vs Gaudry S_4, C_4	St	E	47.5–48.4	$\mathbb{F}_{269^4}$
`gaudry_quartic` solve	St	E	1.9–6.4	$d = 4..6$
One-target online				
Compact-orbit IC vs strong rho	On	E	13.94	$n = 61$, [9.81, 19.13], 64 targets
Accounting (costs that went up when priced)				
F4 word XORs priced	Cold	A	1/25.2	16.75 → 421.4, K13
F6-IC batched geometry priced	Cold	A	1/64.6–1/1,353	$m = 13, 19$

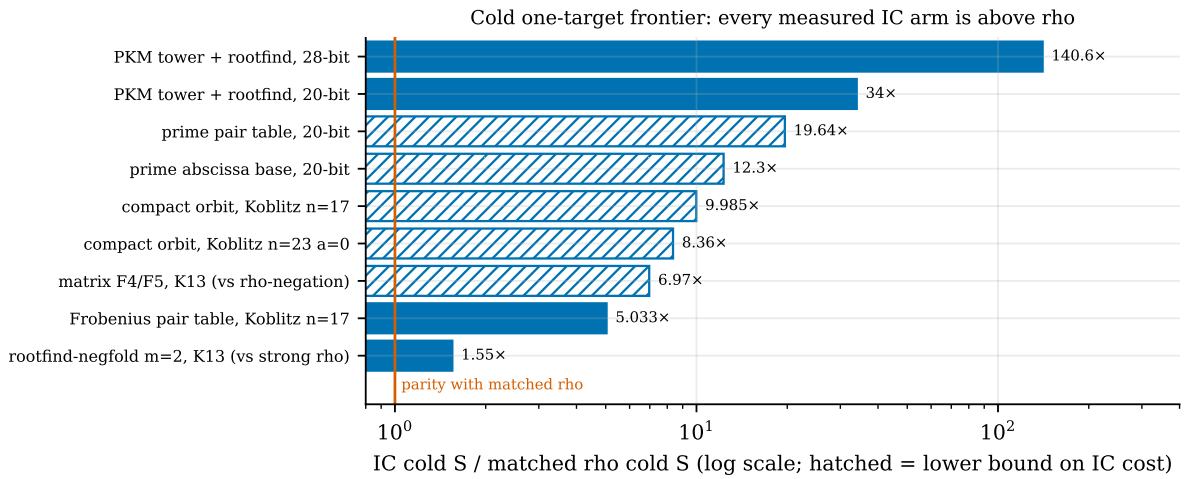


Figure 7: Cold one-target S of index-calculus candidates over their matched rho arm (each study's own estimator; curves differ, so bars are not ranked against each other). Hatched: lower bound on IC cost (unpriced native or solver work). Sources: `research/ecbench_frontier_20261013/`, `research/ecbench_rootfinding_assessment_20261012/`, `research/pkm_tower_ecbench_20261013/`, `research/ecbench_rootfind_faster_20261014/`.

signed-Frobenius rho at $n = 13$; the closest lower-bound arms are 1.08× (compact orbit with a counted Frobenius table) and 6.97× (matrix F4/F5 against rho-negation at K13). Prime-field arms are 12–239× rho and diverge with size.

Online, after retained set-up. One admitted point: compact-orbit IC at Koblitz $n = 61$, 13.9× $[9.8, 19.1]$ over strong rho on 64 unseen public targets, set-up of about 28 s excluded (cold median 83.4× rho). The larger $n = 71, 73, 83$ observations remain exploratory.

Stage. Moved substantially: the decomposition oracle is two to five orders of magnitude cheaper than at the start of the programme on the workloads where root finding applies, the $\mathbb{F}_p$ F4 is at msolve parity, the Boolean F4 family is one to two orders of magnitude cheaper in counted matrix work, and the $k = 4$ extension-field solve is 48× cheaper. None of this, by itself, moves the cold frontier: the stages it leaves untouched—relation yield, factor-base size, linear algebra, table set-up—now dominate.

10.2 What would move the cold frontier

From the frontier synthesis and the problem records: (i) register the torsion-symmetrised bases and the $n = 83$ dense-root engine in ecbench under immutable method identities and emit cold and online records; (ii) resolve the inherited-F4 cross-host counter drift before using that counter in a bound; (iii) run a four-size native challenge against the standing rho bound—two sizes cannot establish an exponent; (iv) obtain an isolated cold session at $n = 83$ with every set-up phase and retained byte charged; (v) price F6-IC's batched geometry with a pinned conversion. A result in the cold domain that beats rho at four sizes would be the first ADVANCE of the programme.

11 Conclusion

Fourteen weeks of work produced three kinds of result. The first is infrastructure: a single unit and harness in which rho, BSGS, the kangaroo and every index-calculus variant are charged the same way on the same one-target workloads, with sealed records, exact replays and admitted intervals; and a ledger that refuses a claim with a missing field. The second is a set of constant-factor improvements, the largest of which—the closed-form roots of S_3—began as one sentence in a manuscript we were asked to check, and ended as a proved identity, an oracle family covering prime and binary fields, two to four summands, CPU and GPU, plain, tower and dense-index bases, and the default decomposition strategy of the codebase. The third is a map of what does not work, which is longer than the list of what does: SAT that does not scale, symmetry that pays at $n = 17$ and costs at $n = 31$, transfers that provably

cannot change decomposition counts, fibre-aware bases whose gap to rho grows as $\#E^{1/6}$, literature attacks that are exhaustive search.

The honest summary is the frontier of §10. On the question the programme was built to answer—does index calculus solve one unseen elliptic-curve target, cold, for less than rho?—the answer at every size we measured is no, and the closest complete arm is 1.55× rho on a Koblitz curve over $\mathbb{F}_{2^{13}}$. With preparation retained, index calculus does beat rho online on Koblitz curves, by an admitted 13.9× at $n = 61$. The stage costs that dominated in September are now small; the stages that remain are relation yield, factor-base size and linear algebra, and that is where the next measurement should look.

A Organising the publication

This document is a programme record: it is complete about what was done and deliberately long. For external submission we recommend splitting it into a short series, each paper self-contained, each citing this record for provenance. Table 3 is the proposed split.

Table 3: Proposed papers. "Ready" means every number exists in a committed artefact; "owed" lists what must be done before submission.

Paper	Content (sections here)	Owed before submission
P1. Root-finding decomposition	§5; extends `papers/rootfinding-decomposition/` with the PDP family, GPU, Kummer tower, ecbench loops and the audit corrections	novelty audit against the literature; a stronger F4 cross-check (Magma/msolve on every row); a GPU run on hardware; $m \geq 3$ prime measurements
P2. ecbench: one unit for every ECDLP method	§3, Def. 1, the corrections list, the isolation rubric, the frontier synthesis	an L3 cross-host replay certificate; resolution of the inherited-F4 counter drift; a four-size bound for each incumbent
P3. Index calculus on Koblitz curves, measured	§7, the compact orbit, the online ladder, the admitted $n = 61$ panel, the ECC2K-130 extrapolation	isolated panels at $n = 71, 73, 83$; reconciliation of the constant C; a cold session at $n = 83$
P4. Symmetry at $k = 4$	§8.4: generic Macaulay solve, D_4 invariants, measured C_4 and rate, modelled crossover	a second prime and more seeds; the degenerate-target cause; an end-to-end measured point instead of the model alone
P5. Negative results	§9, the lit2026 negatives, SAT scaling, symmetrised bases, fibre-aware IC; Lemma 1	literature check of Lemma 1's novelty; the conductor-gap yield experiment (E6)

Suggested outline for each paper. (1) The question, stated as a boundary in the sense of Definition 1. (2) The mathematics, with proofs or certificates. (3) The implementation and its fail-closed verification. (4) Results on the stated boundary first, stage results second, each with unit, sample, interval and artefact. (5) What the result does not establish. (6) Reproduction: commit, binary hash, spec, session id, replay certificate.

Venue. P1, P3 and P4 fit a cryptanalysis venue (an IACR conference or journal; ePrint first, using the repository's `iacr-eprint` workflow); P2 fits an artefact or experimental methodology track; P5 can be a short ePrint note or a section of P2. Every paper must be de-anonymised (`make deanonymize`) and pass `make check` before posting.

B Artefact index

Paths are relative to the root of the `crypto` repository. Each study directory holds its protocol, raw records, run reports with graphs and a PDF; ecbench studies also hold sealed sessions and audit receipts.

Measurement
docs/ecbench/README.md; src/cryptanalysis/ecbench/; docs/ic/BOUNDARY_TARGETS.md; docs/ic/boundary_targets.json; docs/ic/PLAN_IC_ACCOUNTING_FIXES_20261007.md; docs/bounds/frontier.json; research/ecbench_frontier_20261013/.

Root finding
RESEARCH_SEMAEV_ROOTFIND_VS_GROEBNER.md; papers/rootfinding-decomposition/; research/ecbench_rootfinding_assessment_20261012/; research/semaev_root_vs_groebner_20261012/; research/semaev_rootfind_nist_20261012/; research/semaev_rootfind_e2e_20261012/; research/pdp_root_finder_20261012/; research/pdp_root_finder_ecbench_20261013/; RESEARCH_GPU_ROOTFIND_DECOMPOSITION.md; research/ecbench_rootfind_engine_20261012/; research/ecbench_rootfind_improve_20261013/; research/ecbench_rootfind_parity_20261013/; research/ecbench_rootfind_faster_20261014/; research/ecbench_rootfind_vs_rho_20261014/; research/ecbench_rootfinding_tuning_20261013/; research/ecbench_rootfinding_arity_20261013/; research/ecbench_rootfinding_batchtable_20261013/; research/ecbench_rootfinding_lanes_20261013/; research/n83_dense_root_20261013/.

Gröbner engines
docs/ic/GROEBNER_F4F5F6_AUDIT_20261010.md; research/f4_f5_f6_cross_family_20261011/; research/inherited_f4_eager_rewrite_20261011/; research/f4_f5_fused_readback_20261012/; research/f4_f5_hash_20261012/; research/f4_f5_dense_packing_20261012/; research/f4_f5_row_build_20261011/; research/f4_fp_msolve_parity_20261011/; research/gb_cas_comparison_20261011/; research/ic_semiregular_law_20261005/; research/f6_ic_ecbench_ladder_20261005/.

Koblitz
RESEARCH_KOBLITZ_INDEX_CALCULUS.md; RESEARCH_ECC2K130_IC_FEASIBILITY.md; docs/ic/COMPACT_ORBIT_PRODUCER.md; research/ecbench_factor_bases_20261013/; research/koblitz_symmetrised_e2e_20260927/.

Prime and extension fields
research/pdp_oracle_speedup_20261011/; research/pdp_split_dreg_20261011/; research/pkm_tower_ecbench_20261013/; research/gaudry_quartic_d4_20261013/; research/gaudry_quartic_speed_20261013/; research/lit2026_speedup_program/; research/lit2026_w12_flagged_claims_20261011/; research/lit2026_w11_generic_engineering_20261011/.

Transfer and isogeny
docs/ic/RESEARCH_TRANSFER_AWARE_IC_20261010.md; docs/ic/RESEARCH_ISOGENY_CONDUCTOR_GAP_20261007.md; RESEARCH_ISOGENY_FIBERS_ECDLP.md; research/ecbench_conductor_gap_20261012/; research/isogeny_conductor_gap_round3_20261011/; research/fiber_ic_scaling_20261011/; research/isogeny_loop_glv_20261012/.

This paper
papers/ecdlp-index-calculus-program/figures/data.json (every plotted value with its source); figures/make_figures.py; browsing copies in benchmarks/20261014-ic-program-paper/.

References

[1] Claus Diem. On the discrete logarithm problem in elliptic curves. *Compositio Mathematica*, 147(1): 75–104, 2011.

[2] Jean-Charles Faugère. A new efficient algorithm for computing Gröbner bases (F4). *Journal of Pure and Applied Algebra*, 139(1–3):61–88, 1999.

[3] Jean-Charles Faugère. A new efficient algorithm for computing Gröbner bases without reduction to zero (F5). In *Proceedings of ISSAC 2002*, pages 75–83. ACM, 2002.

[4] Jean-Charles Faugère, Pierrick Gaudry, Louise Huot, and Guénaël Renault. Using symmetries in the index calculus for elliptic curves discrete logarithm. *Journal of Cryptology*, 27(4):595–635, 2014.

[5] Steven D. Galbraith, Robert Granger, Simon-Philipp Merz, and Christophe Petit. On index calculus algorithms for subfield curves. In *Selected Areas in Cryptography – SAC 2020*, volume 12804 of *LNCS*. Springer, 2021. Cryptology ePrint Archive, Paper 2020/1315.

[6] Robert P. Gallant, Robert J. Lambert, and Scott A. Vanstone. Improving the parallelized Pollard lambda search on anomalous binary curves. *Mathematics of Computation*, 69(232):1699–1705, 2000.

[7] Pierrick Gaudry. Index calculus for abelian varieties of small dimension and the elliptic curve discrete logarithm problem. *Journal of Symbolic Computation*, 44(12):1690–1702, 2009.

[8] David Jao, Stephen D. Miller, and Ramarathnam Venkatesan. Do all elliptic curves of the same order have the same difficulty of discrete log? In *Advances in Cryptology – ASIACRYPT 2005*, volume 3788 of *LNCS*, pages 21–40. Springer, 2005.

[9] Christophe Petit, Michiel Kosters, and Ange Messeng. Algebraic approaches for the elliptic curve discrete logarithm problem over prime fields. In *Public-Key Cryptography – PKC 2016*, volume 9615 of *LNCS*, pages 3–18. Springer, 2016.

[10] John M. Pollard. Monte Carlo methods for index computation (mod p). *Mathematics of Computation*, 32(143):918–924, 1978.

[11] Igor Semaev. Summation polynomials and the discrete logarithm problem on elliptic curves. Cryptology ePrint Archive, Paper 2004/031, 2004.

[12] Monika Trimoska, Sorina Ionica, and Gilles Dequen. Parity (xor) reasoning for the index calculus attack. In *Principles and Practice of Constraint Programming – CP 2020*, volume 12333 of *LNCS*, pages 774–790. Springer, 2020.

[13] Paul C. van Oorschot and Michael J. Wiener. Parallel collision search with cryptanalytic applications. *Journal of Cryptology*, 12(1):1–28, 1999.

[14] Michael J. Wiener and Robert J. Zuccherato. Faster attacks on elliptic curve cryptosystems. In *Selected Areas in Cryptography – SAC 1998*, volume 1556 of *LNCS*, pages 190–200. Springer, 1999.